

Cellulose Aerogels and BNC in Automotive Interiors: Acoustic & Thermal Applications

Background and Material Properties

Cellulose-based aerogels – including **bacterial nanocellulose (BNC) aerogels** – are ultra-light, porous foams derived from plant fiber or microbial cellulose. They combine **low density and high porosity** with the renewability of cellulose, yielding excellent thermal insulation and promising acoustic absorption capabilities. Lab studies have shown that dried BNC aerogels can achieve **thermal conductivities as low as ~0.025 W/m·K** (on par with air) and maintain **high sound absorption coefficients over 200–5000 Hz** ¹. In other words, a well-designed cellulose aerogel can serve as an effective **heat and sound insulator**, potentially rivaling traditional foams used for noise, vibration, and harshness (NVH) control. The appeal is clear: such aerogels are lightweight, biodegradable, and can be made from **sustainable resources (wood pulp, plant fibers, or bacterial fermentation)**, aligning with the auto industry's push for greener materials.

Despite these advantages, **cellulose aerogels remain an emerging technology**. Producing them at scale is non-trivial – the gels must be carefully dried (often via supercritical or freeze-drying processes) to retain their porous structure. Bacterial nanocellulose, in particular, has had **limited commercial availability**, typically grown as small gel sheets by niche suppliers ². These production challenges have so far kept cellulose aerogels mostly in the realm of research labs and prototypes. The question, then, is how far this technology has penetrated automotive interior applications – especially as **acoustic liners (headliners, door insulators, floor mats)** or **thermal insulators** – and whether any automakers or suppliers have tried them in practice.

Research on Acoustic & Thermal Performance

A substantial body of research in the past decade has explored cellulose and BNC aerogels for insulation. **Academic papers and lab experiments** consistently report that these bio-aerogels can be tailored for NVH purposes. For example, Revin *et al.* (2019) produced BNC aerogels of various densities and found they had **noise absorption coefficients approaching those of commercial acoustic foams**, with strong absorption across a broad frequency range (up to 5 kHz) ¹. Thermal stability of pure cellulose aerogels is moderate (onset of degradation around 230 °C ¹), but sufficient for interior applications, and their flame retardancy can be improved via additives (e.g. silica or intumescent coatings, as explored in other studies).

Crucially, **hybrid aerogels** that combine cellulose with other materials have shown even better performance. **Silica-cellulose hybrid aerogels**, for instance, leverage silica's fire resistance and cellulose's flexibility. Researchers report that these hybrids exhibit **excellent acoustic and thermal insulation properties** – in one study, composite aerogels made from recycled cellulose fibers and a silica precursor had sound absorption coefficients (SAC) in the 0.4–0.5 range (higher is better for absorption) across mid-range frequencies ³. This was better than the SAC ≈ 0.3 of pure cellulose aerogel under the same conditions,

indicating a synergistic effect ⁴. The hybrids were also noted to be more mechanically robust than brittle pure-silica aerogels ³. Another approach has reinforced cellulose aerogels with inorganic or polymer phases (e.g. cellulose-**silica aerogel blankets** or cellulose-polyvinyl alcohol foams) to improve durability in vibration environments ⁵ ⁶. All these developments suggest that, on a **performance basis**, cellulose-based aerogels can meet or even exceed the acoustic/thermal requirements for vehicle interior liners – at least in controlled tests.

In parallel, **patent literature** reflects interest in nanocellulose foams for insulation. For example, a 2022 U.S. patent application describes a method to make cellulose nanofibril aerogel foams with crosslinkers, aiming for materials that could be used similarly to petro-foam insulators ¹. While the patent doesn't explicitly limit to automotive, it underscores a push to scale up **biobased aerogel production**. Another publication by Mroszczok *et al.* (2017) demonstrated **cellulose aerogel fibers** used as a thermal wrap for car engines, to reduce heat loss in hybrids ¹. This concept of *drapable aerogel mats* around engine components showed improved heat retention, hinting that aerogel-like materials could find roles beyond the passenger cabin (e.g. under-hood or battery insulation). In short, the **scientific groundwork** for using cellulose aerogels in vehicles is being laid, with positive results for both acoustic damping and thermal management.

Prototypes and Concept Car Applications

実際の車両への適用例はまだ少ないですが、いくつかの **概念実証** が世界各地で行われています (there are a few proof-of-concept examples globally, though full production use is rare). One high-profile example is Japan's **Nano Cellulose Vehicle (NCV)**, a concept car unveiled at the 2019 Tokyo Motor Show.



The NCV concept car (Japan, 2019) showcased extensive use of cellulose nanofiber materials throughout its body and interior. This supercar-styled prototype was the product of a government-industry-academic project involving 22 entities. About **13 parts of the NCV, including interior door panels, trim and floorboard components, were made with cellulose nanofiber (CNF) composites**, instead of conventional plastic ⁷. Toyota Boshoku (a major Tier-1 supplier) led the development of the CNF-based door trim, building on its experience with natural fibers (like kenaf) in interiors ⁸. The CNF material was combined with resins to form structural parts and was also used in semi-transparent roof panels and exterior bodywork. Overall,

NCV achieved a ~10–16% weight reduction in those components versus standard materials ⁹, demonstrating a significant **lightweighting** benefit. While the NCV's emphasis was weight and sustainability, acoustic and thermal aspects were considered as well – project leaders noted that CNF composites have **low thermal expansion and good stability under heat** ¹⁰ ¹¹, relevant for maintaining fit and finish in the cabin. They also indicated that the material could be processed in various forms: it “**can be injection-molded or formed into foams just like petrochemical plastics**” ¹². This implies that a CNF **foam liner** for a headliner or door could be made using conventional equipment, an enticing prospect for future interiors. The NCV remains a concept car, but starting in 2020 some Japanese OEMs planned limited use of CNF-reinforced plastics in production (e.g. in door panels) ¹³ – a stepping stone toward more ambitious uses like aerogel insulation.

Elsewhere, we do not find reports of entire car interiors using cellulose *aerogel* liners, but smaller trials have occurred. For instance, Ford Motor Company has experimented with **cellulose-based acoustic insulation** in a more traditional form: recycled cotton fiber. The **2012 Ford Focus** incorporated *recycled denim and cotton* as a sound absorber in the carpet backing and other hidden insulator areas ¹⁴. Ford calculated that each Focus used the equivalent of ~2 pairs of discarded jeans for soundproofing, diverting textile waste from landfills and reducing reliance on new fiberglass or foam. This wasn't an aerogel (just a dense fiber mat), but it demonstrated that cellulose fibers can effectively dampen noise in production cars without compromising NVH performance. In the EV realm, Nissan has used recycled blue jeans for sound insulation in the Leaf, and other brands have similar programs ¹⁵. These early adopters of **bio-based acoustic insulation** show that automakers are willing to replace conventional materials with cellulose-derived ones when cost and performance align.

From the **supplier side**, a notable development is Autoneum's **Flexi-Loft** technology – essentially a *foamed felt* made of recycled cotton and other fibers – launched in the early 2020s as an alternative to polyurethane foams. Autoneum (a Swiss Tier-1 specializing in NVH products) reports that Flexi-Loft, composed of **at least 50% recycled cotton fibers**, is already being used in **carpets, headliners, and dash insulators** to cut weight and improve sustainability ¹⁶ ¹⁷. Because it's a fibrous mat (not aerogel), its density is higher than an aerogel's, but it is **lighter and more flexible than traditional felt or foam**. The success of such products indicates that automakers are open to **natural fiber composites for NVH**. It's a short leap in concept from a cotton-fiber mat to a cellulose aerogel panel – the latter could provide even greater weight reduction if manufacturing issues are solved. However, as of 2025, we haven't seen a supplier announce a true **aerogel** (with supercritical-dried type structure) in a production vehicle. The implementations are closer to nonwoven mats or foam-fiber blends, which are easier to mass-produce. In summary, **cellulose aerogel use in cars remains at the prototype stage**, but related technologies (natural fiber felts, CNF-reinforced plastics) are entering the market, validating the general idea.

Adjacent Materials and Comparison

To gauge how *novel* cellulose aerogels are in automotive interiors, it helps to compare with **analogous materials**: some well-established, others experimental.

- **Melamine Foam (Basotect)** – *Status: Widely adopted*. This open-cell melamine-resin foam has been a mainstay for automotive acoustic insulation for years. It is used in **headliners, roof trim, and engine bay liners** for its excellent sound absorption, light weight, and inherent flame resistance (meeting strict fire safety standards) ¹⁸. For example, Basotect® foam panels can be found behind headliner fabrics to dampen cabin noise and in dash insulators to absorb engine sound. Its thermal

insulating value is a bonus in moderating roof heat. Any new material like a cellulose aerogel must compete with melamine foam's combination of properties and cost. Notably, melamine foam is also a **low-density porous material** (~9 kg/m³), though typically heavier than the lightest aerogels. Cellulose aerogels have potential to be even lighter and *greener*, but **melamine sets a high benchmark** for acoustic performance and is a mature, cost-optimized product.

- **Polyurethane and Polyester Foams** – *Status: Widely adopted.* Besides melamine, a lot of vehicle NVH padding uses polyurethane foam (e.g. the backing of carpets, seat padding with acoustic damping function, etc.) or PET fiber pads. These are petroleum-based and not as lightweight as aerogels, but they are inexpensive and easily molded. Many carmakers are actively looking to replace or reduce PU foams due to sustainability goals, which is why natural fiber solutions are attractive. **Cellulose aerogels, if made cost-effective, could replace some PU foam in insulation roles**, offering weight reduction and eliminating fossil-based polymers. At present, **no commercial car uses a cellulose aerogel in place of PU foam**, but the door is open for future trials as automakers phase in bio-based content (for instance, some manufacturers have targets for 20–25% renewable/recycled material in interiors by mid-decade ¹⁹ ²⁰).
- **Natural Fiber Mats (Kenaf, Flax, etc.)** – *Status: Commercial use in specific components.* Natural fiber-reinforced plastics have been used in door panels, package trays, and headliner substrates by brands like BMW, Mercedes, Audi, and others. These typically consist of **flax, hemp, kenaf, or wood fibers mixed with polypropylene** and compression-molded into a lightweight panel. They offer decent sound absorption (the hollow plant fibers damp sound) and weight savings (~20–30% lighter than all-plastic) while being largely renewable. For example, Mercedes-Benz's concept and production models have used flax-based composites for console and door parts, and Audi has used kenaf fiber in some interior linings ²¹ ⁸ . The presence of these materials shows that automakers are comfortable with **cellulose in a composite form**. However, these are not aerogels – the fibers are still relatively dense when packed in a composite. A cellulose aerogel liner could outperform such mats by having an order of magnitude lower density and a more open pore structure for sound dissipation. In essence, current natural fiber components are a halfway point: they prove the **NVH effectiveness of cellulose** (even in coarse fiber form), hinting that a more engineered form like an aerogel could be the next leap if practical.
- **Mycelium-based Foams** – *Status: Experimental/prototype.* *Mycelium*, the thread-like root network of fungi, can grow into a foam-like biomass that is being explored as a sustainable foam alternative. In lab tests, mycelium composite panels have shown **sound absorption comparable to standard acoustic boards**, and even superior in some metrics. Notably, researchers Gezer and Kuştas (2022) measured mycelium-insulation boards (made from fungus-grown wheat straw) versus conventional materials: the mycelium boards achieved *perceived road noise levels of ~45–60 dBA*, outperforming a polyurethane foam board (64 dBA) under the same test conditions ²² . The mycelium's natural damping was especially strong at low frequencies (<1500 Hz), even better than cork or commercial ceiling tiles ²³ . These promising results have caught industry attention. In 2023–2024, automotive designers flirted with mycelium in niche areas – for example, General Motors' **Cadillac Celestiq SOLLEI concept** incorporated a mycelium-derived faux leather in its interior trim as a novelty ²⁴ , and Kia's EV2 concept (shown at Milan Design Week 2025) reportedly used mycelium and hemp for certain cabin parts. **However, no production vehicle yet uses mycelium foam for acoustic lining**; the technology is still young and focused more on construction and packaging industries. Mycelium foams and cellulose aerogels are conceptually similar (both are low-density biological foams), so the

tentative steps with mycelium underscore that the auto industry is testing the waters with unconventional biomaterials. Their adoption will hinge on meeting fire safety and durability specs, as well as scaling production.

- **Cellulose-Silica and Other Hybrid Aerogels** – *Status: Research stage.* As mentioned, blending cellulose with inorganic components (silica, clay, etc.) can yield a hybrid aerogel with improved fire resistance and perhaps better acoustic profiles. One recent study achieved nearly **99% sound absorption ($\alpha \approx 0.99$) at certain frequencies** with a silica-cellulose aerogel composite derived from rice straw ²⁵ ⁶. Such extreme performance is lab-based, but it highlights the potential if these materials could be molded into car-ready parts. There have also been explorations of **cellulose aerogels reinforced with graphene or carbon nanotubes** for electromagnetic shielding and of **rubber-infused aerogels** (e.g. using recycled tire rubber with silica aerogel) for flexible insulation panels ²⁶ – these might find niche automotive uses (for example, an aerogel that also provides EMI shielding for EV electronics). These hybrid approaches remain *novel concepts* with no automotive deployments yet, but they contribute to a knowledge base that could make future cellulose aerogel liners more functional and easier to integrate (e.g. a rubber-cellulose aerogel might double as a sound absorber and a vibration damper).

In comparing all these materials, it's evident that **cellulose-based aerogels are among the most novel** – they are not in production cars today, whereas melamine foams and natural fiber mats are. Even mycelium, a newcomer, has seen concept car use, albeit in non-critical roles. The aerogel form of cellulose is still *emerging technology* in 2025. Its novelty also means it could be a breakthrough: if it achieves equal or better NVH performance at lower weight and with eco-friendly credentials, it would be a compelling replacement for incumbents.

Current Status and Outlook

After surveying literature and industry reports across the US, EU, and Japan, we find **no evidence of a mass-produced vehicle using cellulose or BNC aerogel as an interior acoustic or thermal liner** as of 2025. The consensus is that this approach remains **innovative and not yet commercialized**. Automakers and their suppliers have certainly shown interest in sustainable NVH materials – using recycled textiles, exploring mycelium, and incorporating natural fibers into composites – but the specific leap to an aerogel (with its demanding fabrication) has been limited to R&D. The **manufacturability** is the key hurdle: supercritical drying or similar processes are expensive and hard to apply to large, thin parts like a headliner. There are also practical considerations like **integration and durability**. An aerogel sheet on its own can be fragile; in an automotive liner it might need a protective facing or to be sandwiched in a composite for handling and longevity. This adds complexity. By contrast, something like Autoneum's fiber felt can be produced with conventional textile equipment and easily molded.

That said, the field is progressing. The NCV concept in Japan suggests that **automotive-grade cellulose materials** can be made and used in complex shapes, hinting that aerogel-like foams could follow. Tier-1 suppliers are actively innovating in this space – for example, Johns Manville and BASF have each developed *enhanced insulation foams* (BASF's Basotect is one, and there are others with polymer-fiber blends) targeting EV NVH, since electric cars have new noise profiles (e.g. high-frequency inverter whine and a lack of engine masking noise) that demand improved sound damping. It's plausible that a **start-up or collaborative project** will soon demonstrate a cellulose aerogel headliner or floor mat in a prototype EV, to claim the

sustainability edge. We have already seen **patents and papers** indicating that continuous-roll production of nanocellulose aerogel insulation is being pursued ²⁷ ²⁸ – a necessary step for automotive adoption.

In conclusion, **cellulose-based and BNC aerogels for automotive interiors are still novel, with usage confined to prototypes and laboratory studies so far.** They have *indeed been tried* in concept form (e.g. the NCV's use of CNF foams, research engine encapsulation blankets, etc.), and those trials show **strong technical potential** – high acoustic absorption and excellent thermal insulation, all with a renewable material ²⁹. However, when placed in the context of commercial automotive materials, they are **not yet widely adopted**. More common today are adjacent solutions like recycled fiber mats or conventional foams, which cellulose aerogels would aim to replace or improve upon. The approach of using cellulose aerogels as cabin noise/heat liners **remains cutting-edge in 2025**, still requiring scale-up and validation in the harsh automotive environment (temperature swings, moisture, long-term vibration). Industry experts seem to view it as a promising avenue for the future – part of the broader trend of replacing synthetics with bio-derived materials – but one that might take another few development cycles to materialize in showrooms. In summary, **the concept is viable and attractive, but its novelty means we're likely only in the early stages of seeing it move from research to real-world automotive applications** ¹³ ³⁰.

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